Online Examination System

Project Documentation

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1. Abstract

The Online Examination System is a web-based application that lets institutions conduct exams digitally. Administrators create exams and question banks, students log in and attempt timed tests, and the system evaluates answers automatically and publishes results instantly. The system removes paper-based work, reduces evaluation time and human error, and makes exams easier to manage and audit.

2. Introduction

Traditional exams need printed papers, invigilation logistics, manual evaluation, and long result turnaround times. An online examination system solves these problems by moving the entire process onto a secure web platform. Students can attempt exams from a computer lab or any approved device, and results are available as soon as the exam ends.

3. Problem Statement

Paper-based exams are slow, costly, and prone to errors in marking and record keeping. Institutions need a system that can create exams quickly, deliver them securely within a fixed time limit, evaluate objective answers automatically, and store results safely for later reference.

4. Objectives

- Provide a secure login for administrators and students.
- Allow admins to create exams, manage subjects, and maintain a question bank.
- Deliver timed multiple-choice exams with automatic submission when time ends.
- Evaluate answers automatically and calculate scores without manual effort.
- Store results and let students view their performance history.
- Reduce paper use, cost, and result-processing time.

5. Existing System vs Proposed System

Aspect	Existing (Paper-based) System	Proposed (Online) System
Question paper	Printed and distributed	Stored digitally, shown on screen
Evaluation	Manual, slow, error-prone	Automatic and instant
Result time	Days or weeks	Immediately after submission
Record keeping	Physical files	Centralized database
Cost	Paper, printing, staff time	Low recurring cost
Security	Risk of paper leaks	Role-based access and controlled delivery

6. Scope

The system covers user management, exam creation, question management, exam attempt, automatic evaluation, and result reporting for objective (multiple-choice) questions. Descriptive-answer evaluation, video proctoring, and mobile apps are outside the current scope and are listed under future enhancements.

7. System Modules

7.1 Admin Module

- Log in securely and manage the dashboard.
- Add, edit, and delete subjects and exams.
- Add, edit, and delete questions with four options and the correct answer.
- Set exam duration, total marks, and passing marks.
- View all students' results and export reports.

7.2 Student Module

- Register and log in.
- View available exams and instructions.
- Attempt the exam with a countdown timer and question navigation.
- Submit manually or have the exam auto-submit at time-out.
- View score, correct and wrong answers, and past results.

7.3 Evaluation Module

- Compares each selected answer with the stored correct answer.
- Calculates total score, percentage, and pass or fail status.
- Saves the result to the database.

8. System Requirements

8.1 Functional Requirements

- Users must authenticate before accessing the system.
- Admins can create and manage exams and questions.
- Students can attempt each exam within its time limit.
- The system must evaluate answers and store results automatically.
- Students can only view their own results.

8.2 Non-Functional Requirements

- **Security:** passwords stored in hashed form, role-based access, session timeout.
- **Performance:** pages load quickly and support many students at once.
- **Reliability:** answers are saved so a refresh or short disconnect does not lose progress.
- **Usability:** simple, clear interface that needs no training.
- **Maintainability:** modular code that is easy to extend.

8.3 Software Requirements

Component	Technology
Front end	HTML5, CSS3, JavaScript (fetch API for back-end calls, exam countdown timer)
Back end	Java with Spring Boot (REST controllers, Spring Security, Spring Data JPA)
Database	MySQL
Scripting and utilities	Python (bulk question import from CSV, result analysis reports)
Build tool	Maven
IDE	Eclipse or VS Code
Server	Embedded Tomcat (included with Spring Boot)
Version control	Git and GitHub

8.4 Hardware Requirements

- Processor: Intel Core i3 or higher
- RAM: 4 GB minimum
- Storage: 500 MB free space
- Internet or LAN connection

9. System Architecture

The system follows a three-tier architecture:

1. **Presentation layer:** HTML, CSS, and JavaScript pages for login, dashboard, and exam screens.
2. **Application layer:** Spring Boot REST APIs that handle authentication, exam logic, timer control, and evaluation.
3. **Data layer:** MySQL database accessed through Spring Data JPA that stores users, exams, questions, and results.

The browser sends a request to the application layer, which reads or writes data in the database and returns the response to the user.

Inside the Spring Boot application, requests flow from controller to service to repository. JavaScript on the front end calls the controllers' REST endpoints and renders the questions and timer. Python runs outside the web application as helper scripts: one imports questions from a CSV file into MySQL, and another generates score and pass-percentage reports from the results table.

10. Process Flow

1. The user opens the login page and enters credentials.
2. The system verifies credentials and checks the role (admin or student).
3. **Admin:** creates exams and questions, then views results.
4. **Student:** selects an exam, reads the instructions, and starts the timer.
5. The student answers questions and submits, or the system auto-submits when time ends.
6. The evaluation module calculates the score and saves the result.
7. The student views the result immediately.

11. Database Design

11.1 Tables

users

Column	Type	Description
user_id	INT (PK, auto increment)	Unique user ID
name	VARCHAR(100)	Full name
email	VARCHAR(100), unique	Login email
password	VARCHAR(255)	Hashed password
role	VARCHAR(20)	admin or student

exams

Column	Type	Description
exam_id	INT (PK)	Unique exam ID
title	VARCHAR(150)	Exam name
subject	VARCHAR(100)	Subject
duration_minutes	INT	Time limit
total_marks	INT	Maximum marks
pass_marks	INT	Minimum marks to pass

questions

Column	Type	Description
question_id	INT (PK)	Unique question ID
exam_id	INT (FK to exams)	Exam it belongs to
question_text	TEXT	The question
option_a to option_d	VARCHAR(255)	Four answer options
correct_option	CHAR(1)	A, B, C, or D
marks	INT	Marks for the question

results

Column	Type	Description
result_id	INT (PK)	Unique result ID
user_id	INT (FK to users)	Student
exam_id	INT (FK to exams)	Exam attempted
score	INT	Marks obtained
percentage	DECIMAL(5,2)	Score percentage
status	VARCHAR(10)	Pass or Fail
attempted_on	DATETIME	Date and time

11.2 Relationships

- One exam has many questions.
- One student can attempt many exams, and one exam can be attempted by many students.
- Each result links one student to one exam.

12. Evaluation Logic

```
score = 0
for each question in the exam:
    if selected_option == correct_option:
        score = score + marks
percentage = (score / total_marks) * 100
status = "Pass" if score >= pass_marks else "Fail"
save score, percentage, status to the results table
```

13. Sample SQL Queries

```
-- Questions for an exam
SELECT * FROM questions WHERE exam_id = 1;

-- Results of a student with exam names
SELECT e.title, r.score, r.percentage, r.status
FROM results r
JOIN exams e ON r.exam_id = e.exam_id
WHERE r.user_id = 5;

-- Average score per exam
SELECT exam_id, AVG(score) AS avg_score
FROM results
GROUP BY exam_id;
```

14. Testing

Test case	Input	Expected result
Valid login	Correct email and password	Redirect to the role's dashboard
Invalid login	Wrong password	Error message shown
Timer expiry	Time reaches zero	Exam auto-submits
All correct answers	Every option correct	Full marks, Pass
No answers	Submit blank	Zero marks, Fail
Unauthorized access	Student opens an admin page	Access denied
Add question	Valid question data	Saved and visible in the exam

15. Advantages

- Instant and accurate results.
- Saves paper, printing, and evaluation time.
- Centralized and searchable records.
- Secure, role-based access.
- Easy to scale to more students and subjects.

16. Limitations

- Supports objective (multiple-choice) questions only.
- Needs reliable power and network during exams.
- No built-in video proctoring, so cheating prevention is limited.

17. Future Enhancements

- Descriptive and coding question types.
- Random question order and question pools to reduce copying.
- Webcam proctoring and tab-switch detection.
- Email notifications and downloadable PDF certificates.
- Analytics dashboards for performance trends.
- Mobile-friendly app.

18. Conclusion

The Online Examination System makes exams faster, fairer, and easier to manage. By automating exam delivery, evaluation, and result storage, it saves time for institutions and gives students immediate feedback. The modular design allows new features to be added in the future.

19. References

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